

Tianjun Zhong

tianjun.zhong@columbia.edu | (470) 855-3939 | [Homepage](#)

EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science

New York, NY

M.S. Computer Science, Track: Machine Learning (Advanced Master's Research)

Sep. 2024 - May 2026

- **GPA:** 3.96/4.00
- **Relevant Courses:** Natural Language Processing (A+), Spoken Language Processing (A), Computer Vision (A), AI (A-)

Emory University, Emory College of Arts and Sciences, Goizueta Business School

Atlanta, GA

Double Major: B.A. Computer Science, B.B.A. Business Administration

Aug. 2020 - May. 2024

- **Overall GPA:** 3.88/4.00, **Computer Science GPA:** 3.97/4.00
- **Awards:** Graduation with Distinction (top 20%), Dean's List Fall 2020, Fall 2021, Spring 2022, Fall 2023, Spring 2024
- **Relevant Courses:** Machine Learning (A), Analysis of Algorithm (A), Data Mining (A)

PUBLICATIONS

[1] Far from the Shallow: Brain-Predictive Reasoning Embedding through Residual Disentanglement

Linyang He*, **Tianjun Zhong***, Richard Antonello, Gavin Mischler, Micah Goldblum, Nima Mesgarani

NeurIPS 2025 [[arXiv](#)] [[OpenReview](#)]

[2] K2-Think: A Parameter-Efficient Reasoning System

Zhoujun Cheng*, Richard Fan*, Shibo Hao*, Taylor W. Killian*, Haonan Li*, Suqi Sun*, Hector Ren, Alexander Moreno, Daqian Zhang, **Tianjun Zhong**, Yuxin Xiong, Yuanzhe Hu, Yutao Xie, Xudong Han, Yuqi Wang, Varad Pimpalkhute, Yonghao Zhuang, Aaryamonvikram Singh, Xuezhi Liang, Anze Xie, Jianshu She, Desai Fan, Chengqian Gao, Liqun Ma, Mikhail Yurochkin, John Maggs, Xuezhe Ma, Guowei He, Zhiting Hu, Zhengzhong Liu*, Eric P. Xing

arXiv preprint [[arXiv](#)] [[Hugging Face](#)] [[GitHub](#)]

* Equal Contribution

RESEARCH EXPERIENCES

Probing the Evolution of Structured Reasoning in Large Language Models

Columbia University Department of Electrical Engineering

New York, NY

Research Assistant, Project Lead (Advisor: Professor Nima Mesgarani)

Nov. 2025 – Present

- Investigating whether and where LLMs internally form structured chains of thought by comparing model layer representations against ground-truth reasoning trees; developing probes and representation analyses to identify reasoning-step-specific layers and understand how multi-step inference emerges; targeting *ACL 2026*. More details will be shared upon publication.

Multi-Expert-Agent System for Software Engineering

Columbia University Department of Computer Science

New York, NY

Research Assistant (Advisor: Professor Baishakhi Ray, Mentor: Yangruibo (Robin) Ding)

Sep. 2025 – Present

- Designing collaborative LLM agents (<14B) to assist developers with complex software engineering tasks, targeting a paper submission to *ICML 2026*; the goal is to achieve GPT-level performance in agentic software engineering with 10 – 100 times lower computational cost and latency. More details will be shared upon publication.

K2-Think: A Parameter-Efficient Reasoning System

UC San Diego Department of Computer Science & Engineering

San Diego, CA

Research Assistant (Advisor: Professor Zhiting Hu)

May 2025 – Sep. 2025

- Employed large-scale post-training techniques for 32B and 72B parameter LLMs to bridge the performance gap between open-source and proprietary reasoning models; engineered and managed distributed training jobs on clusters of up to 512 GPUs using SLURM, configuring sequence parallelism, packing, and advanced learning rate scheduling for efficiency.
- Developed automated data curation and evaluation pipelines, authoring tools for model distillation and LLM-powered dataset labeling, leveraging PyTorch, DeepSpeed, and Hugging Face Transformers.
- Achieved state-of-the-art 86.26 on AIME 2024 and a competitive 60.90 on LiveCodeBench via supervised fine-tuning (SFT) and reinforcement learning (RL), outperforming open-source models at comparable scale; the resulting model was released publicly as *K2-Think* on Hugging Face, with the complete recipe—from data to training pipelines—made openly accessible through the project's technical report and GitHub repository.

Brain-Predictive Reasoning Embedding through Residual Disentanglement

Columbia University Department of Electrical Engineering

New York, NY

Research Assistant, Project Co-lead (Advisor: Professor Nima Mesgarani)

Jan. 2025 – Aug. 2025

- Developed a residual disentanglement framework to isolate distinct linguistic representations (lexicon, syntax, meaning, reasoning) from LLM activations using minimal-pair diagnostic datasets (BLiMP, COMPS, ProntoQA, WinoGrande); employed layer-wise probing and iterative ridge regression to extract residual embeddings that are linearly unpredictable from lower-level features, achieving effective feature separation.
- Validated the framework by showing that each residual embedding selectively captures its intended feature through targeted diagnostic tasks, and mapped reasoning embeddings to human ECoG recordings, revealing stronger correlation with high-level reasoning cortical regions than classical language areas and later neural activation peaks.
- Provided the first evidence of brain-relevant reasoning representations in LLMs, advancing interpretability and enabling more precise analysis of model cognition; framework offers a generalizable methodology for probing and improving reasoning ability in large models, with paper accepted to *NeurIPS 2025*.

Retrieval-Augmented Generation with Unsupervised Learning Enabling Hallucination Mitigation

Columbia University Department of Computer Science

New York, NY

Research Assistant (Advisor: Professor Vishal Misra)

Sep. 2024 - Dec. 2024

- Built a Retrieval-Augmented Generation (RAG) system to ease academic-advising bottlenecks, reducing faculty workload and improving student access; deployed in Columbia's CS Department with automatic escalation of out-of-domain queries.
- Designed an unsupervised out-of-domain detector by modeling knowledge bases as hyper-ellipsoid clusters, using agglomerative clustering and statistical tests to flag queries beyond knowledge boundaries.
- Achieved a 78.8% rejection rate for hallucination-prone queries, significantly improving factual reliability in responses.

WORK EXPERIENCES

TripleE Group, Department of Product R&D

Shenzhen, China

AI Engineering Intern

May 2024 - Aug 2024

- Designed and implemented three original algorithms to improve RAG pipelines: (1) enhanced query-document matching, (2) dynamic top-*k* retrieval, and (3) optimized text splitting for vector database construction.
- Evaluated three state-of-the-art embedding models for retrieval accuracy, integrated a reranker for relevance ranking, and achieved 98.45% top-1 accuracy on synthetic high-difficulty queries containing ambiguous concepts and expressions.
- Developed efficient web scraping methods to collect over 1.27M tokens of text, constructing 23 FAISS vector databases for downstream retrieval tasks.

Tencent, Department of IDC Platform

Shenzhen, China

Software Engineering Intern

May 2023 - Jun 2023

- Developed Python automation tools to streamline daily operations for the IDC Platform team, reducing manual workload and redundancy by 90%.
- Built programs to monitor 30+ data centers and automatically generate two daily operational reports.
- Integrated solutions with Git platform and cloud database, creating reusable Python functions for team-wide adoption.

TEACHING & CONTEST

Emory University, Goizueta Business School

Atlanta, GA

Teaching Assistant (Course: Applied Data Analytics with Coding)

Jan 2024 - May 2024

- Mentored 122 students through personalized guidance on Python and SQL during weekly two-hour office hours, explaining complex concepts in clear language to a diverse student body.
- Delivered detailed, constructive feedback on assignments, projects, and exams, helping students improve both technical skills and conceptual understanding.

International Collegiate Programming Contest (ICPC), Team Leader, Contestant

Sep 2022 - Feb 2024

- Ranked 3rd among 103 teams in 2023 ICPC Southeast USA Regional Contest, 19th among 110 teams in 2022 ICPC
- Exercised Java and Python skills in dynamic programming, data structures, greedy algorithm in weekly two-hour practices

SKILLS

Programming: Python (Transformers, PyTorch, TensorFlow, Scikit-learn, NumPy), Java, SQL, C, R

Language: English (primary language of instruction for 10 years, professional fluency, GRE 335), Mandarin (native speaker), Cantonese (intermediate), Japanese (elementary)